

```
Run uv add --upgrade marimo or python -m pip install --upgrade marimo to upgrade.
```

Create or edit notebooks in your browser

Macintosh HD > Usr > van > Doc > Doc > Mat > Gith > 03_calculo > 03_calculo_ejercicio_minimizacion.ipynb

Nombre	Fecha de modificación	Tamaño	Clase
03_calculo_diferenciacion_tipos.ipynb	24/03/26, 5:54 p.m.	232 KB	Documento
03_calculo_ejercicio_minimizacion.ipynb	02/09/25, 9:15 a.m.	189 KB	Documento
03_calculo_intro.ipynb	28/08/25, 9:22 a.m.	756 KB	Documento
03_calculo_minimizacion_costos.ipynb	02/09/25, 12:55 p.m.	202 KB	Documento
▸ gradiente_descendiente	30/04/26, 10:54 a.m.	--	Carpeta
gradiente_descendiente.zip	04/09/25, 9:09 a.m.	2.1 MB	Archivo ZIP
▸ mlp	03/09/26, 7:25 a.m.	--	Carpeta
PCA.ipynb	19/03/26, 11:59 a.m.	80 KB	Documento
prices.csv	01/09/25, 10:21 p.m.	933 bytes	Documento CSV

```
Are you sure you want to quit? (y/N): y
```

Thanks for using marimo! 🌊🌿

→ Deep_Learning_CUGDL git:(main) x

Resumen Reg. Lineal Simple (Regression)

$y \in \mathbb{R}$

Modelo

$$\hat{y} = \beta_0 + \beta_1 x$$

Datos

$$y, x$$

Función de Costo/Erro

$$J(\vec{\beta}) = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

Análisis
optimización

$$\left. \begin{aligned} \frac{\partial J}{\partial \beta_0} &= 0 \\ \frac{\partial J}{\partial \beta_1} &= 0 \end{aligned} \right\}$$

Cond.
de
minimización

Sist. de eqs.

$$\frac{1}{n} \sum y_i = \beta_0 + \beta_1 \frac{1}{n} \sum x_i \quad \dots (1)$$

$$\frac{1}{n} \sum x_i y_i = \beta_0 \sum x_i + \beta_1 \frac{1}{n} \sum x_i^2 \quad \dots (2)$$

Solución
analítica

$$\beta_1 = \frac{\sigma_{xy}}{\sigma_{x^2}}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

"mejor ajuste"

$\min(J)$

• Regression Linear Múltiple

Modelo

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_m x_m$$

Función de Costo

$$J(\vec{\beta}) = \frac{1}{n} \sum (\hat{y}_i - y_i)^2$$

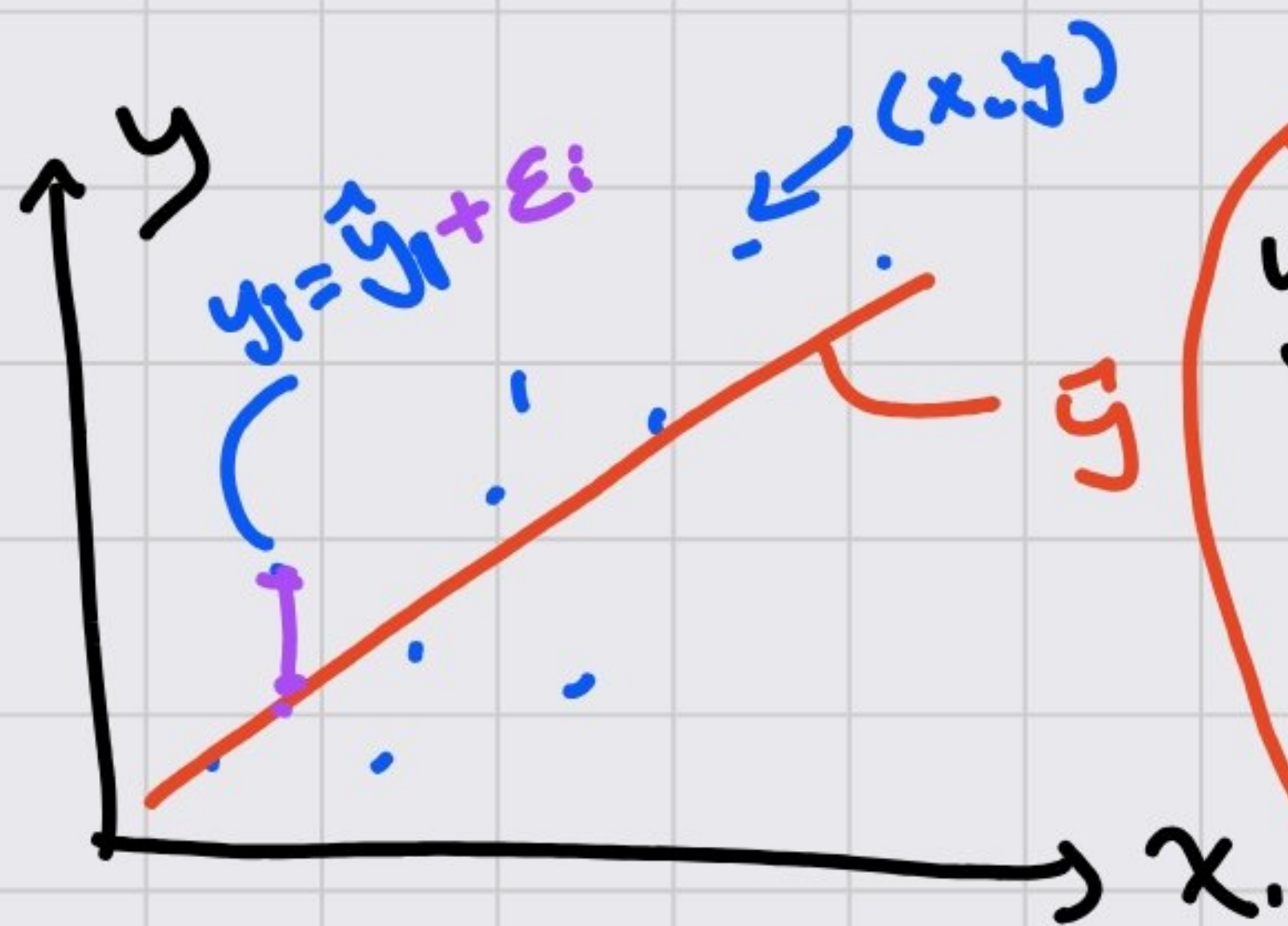
→ residuos $\epsilon_i = \hat{y}_i - y_i$

optimizar J

DATOS

$$\underline{X} = \{x_1, x_2, \dots, x_m\}, y$$

$$\frac{\partial J}{\partial \beta_0} = 0, \frac{\partial J}{\partial \beta_1} = 0, \\ \frac{\partial J}{\partial \beta_2} = 0, \dots, \frac{\partial J}{\partial \beta_m} = 0$$



$$y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \beta_3 x_{13} + \dots + \beta_m x_{1m} + \epsilon_1$$

$$y_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \beta_3 x_{23} + \dots + \beta_m x_{2m} + \epsilon_2$$

$$\vdots$$

$$y_n = \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \dots + \beta_m x_{nm} + \epsilon_n$$

$$\hat{y} = \beta_0 + \beta_1 x_1 + \dots + \beta_m x_m$$

↑
of vars

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} \underbrace{1}_{\text{intercept}} & x_{11} & x_{12} & \dots & x_{1m} \\ 1 & x_{21} & x_{22} & \dots & x_{2m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nm} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

$Y = X B + \varepsilon$

TARGET $\uparrow$ $\uparrow$ characteristics $\uparrow$ params. $\nwarrow$ residuals

$$Y = XB + \varepsilon$$

f. Costo

$$J(\beta) = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

$$= \frac{1}{n} \sum \varepsilon_i^2$$

$$\Rightarrow \varepsilon = Y - XB$$

$$\Rightarrow J = \varepsilon^T \varepsilon$$

$$A \rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad \text{2x3}$$

$$A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}, \quad \text{3x2}$$

$$= (Y^T - X^T B^T) (Y - XB)$$

$$= Y^T Y - Y^T \underbrace{XB}_{\beta_0 + \beta_1 x_1 + \beta_2 x_2} - \underbrace{X^T B^T}_Y Y + \underbrace{X^T B^T}_{\beta_0 + \beta_1 x_1 + \beta_2 x_2} XB$$

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2$$

$$\mathcal{L} = Y^T Y - Y^T X \beta - X^T \beta^T Y + X^T \beta^T X \beta$$

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial \beta} = 0 = \frac{\partial}{\partial \beta} (Y^T Y) - \frac{\partial}{\partial \beta} (Y^T X \beta) - \frac{\partial}{\partial \beta} (X^T \beta^T Y) + \frac{\partial}{\partial \beta} (X^T \beta^T X \beta)$$

$\frac{d}{d\beta} (\beta^2) = 2\beta$

$$= 0 + Y^T X - X^T Y + 2X^T X \beta = 0$$

$\beta^T \beta$

$$a \cdot \frac{1}{a} = 1$$

$$A \cdot A^{-1} = I$$

$$Y^T X = X^T Y \Rightarrow -2X^T Y + 2X^T X \beta = 0$$

$$\Rightarrow 2(-X^T Y + X^T X \beta) = 0$$

$\Rightarrow$

Form matrix

$$\Rightarrow -X^T Y + X^T X \beta = 0$$

parameters solution

$$\Rightarrow (X^T X) \beta = X^T Y$$

$$\beta = (X^T X)^{-1} X^T Y$$

$$d\mathbf{A} = 0 \text{ (for constant } \mathbf{A}\text{)}$$

$$d(\alpha \mathbf{X}) = \alpha d\mathbf{X}$$

$$d(\mathbf{X} + \mathbf{Y}) = d\mathbf{X} + d\mathbf{Y}$$

$$d(\text{tr}(\mathbf{X})) = \text{tr}(d\mathbf{X})$$

$$d(\mathbf{X}\mathbf{Y}) = (d\mathbf{X})\mathbf{Y} + \mathbf{X}d\mathbf{Y}$$

$$d(\mathbf{X} \otimes \mathbf{Y}) = (d\mathbf{X}) \otimes \mathbf{Y} + \mathbf{X} \otimes d\mathbf{Y} \text{ (see section 2)}$$

$$d(\mathbf{X} \circ \mathbf{Y}) = (d\mathbf{X}) \circ \mathbf{Y} + \mathbf{X} \circ d\mathbf{Y} \text{ (see section 5)}$$

$$d\mathbf{X}^{-1} = -\mathbf{X}^{-1}(d\mathbf{X})\mathbf{X}^{-1}$$

$$d\|\mathbf{X}\| = \|\mathbf{X}\|^{-1}(\mathbf{X}^{-1}d\mathbf{X})$$

Soluciones Analíticas

Reg. Lineal Mult

$$\beta = (X^T X)^{-1} X^T Y$$

Reg. Lineal Simple

$$\beta_1 = \frac{\sigma_{xy}}{\sigma_x^2}, \beta_0 = \bar{y} - \beta_1 \bar{x}$$

Parámetros de ajuste

```
print(f"beta_0 : {model.intercept_}")
```

```
print(f"beta_1 : {model.coef_[0]}")
```

```
print(f"beta_2 : {model.coef_[1]}")
```

... beta_0 : 9.21935350605318

beta_1 : -0.7321610332915767

beta_2 : -0.0028107673413597513

$$\text{mpg}_{\log} = 9.21 + (-0.73) \cdot \text{weight}_{\log} + (-0.002) \cdot \text{horsepower}$$



Start coding or generate with AI.

Tensor

Escalar
3

Orden
0

Vector
(3, 4, 5)
 $i \rightarrow$ position.

Orden
1

Matrix
 $i \rightarrow$
 $j \rightarrow$
 $\begin{pmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$

Orden
2